Morphological Classification of Galaxies in HST and JWST Images Using Classical DIP

Weekly Progress Report — Week 8

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ECE 501 – Digital Image Processing Course Project

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1 Purpose and Scope

Week 7 closed out formal rule-based classification on Stephan's Quintet alone and flagged two forward actions: revise the decision rule, and scale the pipeline to additional Hickson Compact Groups. Week 8 is the scale-up step, and most of its effort was consumed by a single, severe data-quality problem that surfaced the moment a second real target was run through the pipeline. The highlights below are a selection of the week's most consequential changes, not an exhaustive log:

1. **Multi-Target Pipeline Generalization**: refactor the single-target star/galaxy-separation, cutout-extraction, and report-generation cells into reusable, target-agnostic functions driven by one registry and one loop, so any target can run through identical code (Section 2).
2. **New Target Onboarding**: bring two new groups, HCG 90 and HCG 16 (HST WFPC2/PC imaging), online under the generalized pipeline, using generic (unnamed) candidate labels per the standing no-name-classifier instruction, while leaving Stephan's Quintet's existing named cross-matching untouched.
3. **HCG 16 Over-Detection Diagnosis and Fix**: HCG 16's WFPC2 mosaic triggered a false-positive detection rate roughly five times every other target's, traced to its data footprint and corrected through an iterative, evidence-driven fix sequence (Section 3) – the bulk of this week's work.
4. **Deferred Issue, Logged Not Fixed**: cosmic-ray / satellite-streak contamination in HCG 16's candidate list was identified but deliberately left unaddressed this week, carried forward as an explicit caveat.

All work this week remains deterministic and rule-based – new per-target parameters are explicit numeric overrides, not fitted or learned values, and no machine learning is used anywhere in the pipeline.

2 Multi-Target Pipeline Generalization

2.1 Methodology

The pre-Week-8 pipeline hard-coded Stephan's Quintet throughout the star/galaxy-separation, cutout-extraction, and results-reporting stages. These were replaced with a single reusable orchestrator, `run_target_pipeline()`, driven by a `TARGETS_CONFIG` registry that holds each target's FITS paths, filters, and (where available) a name-matching catalog. A target is activated only if its FITS files actually resolve on disk; an unavailable target is skipped with a clear message rather than failing the run. Stephan's Quintet keeps its existing `SQ_CATALOG`-based WCS name matching; HCG 90 and HCG 16 use generic `Obj1`, `Obj2`, ... labels ranked by area, since no name-matching catalog was built for them.

2.2 Outcome

The driver loop now produces one combined feature table, `df_cas_all`, pooling every active target's galaxies under identical column definitions, and one 4-panel (C/A/S/S_gran) distribution figure per target rather than a single-target plot. This is the structural prerequisite for both the broader benchmarking Week 7 called for and the further scale-up carried out in later weeks.

3 HCG 16 Over-Detection: Root Cause and Fix Sequence

3.1 Diagnosis

On the first real run, HCG 16 flagged roughly 9–10% of pixels as statistically significant even at a raised 25σ threshold – against a well-behaved $<2\%$ for every other target. Comparing HCG 16's raw FITS array directly against HCG 90's as a working baseline showed the actual cause: HCG 16's WFPC2/PC file is 53.90% non-finite (a non-rectangular 4-chip mosaic footprint), versus HCG 90's clean 7.26% edge rim. The existing scalar NaN-fill made any local background box landing inside that region perfectly flat, driving its local noise estimate to ≈ 0 and

making any adjacent real pixel trivially "significant" at any threshold – which is exactly why raising the threshold alone barely moved the result.

3.2 Fix Sequence and Results

Table 1: Successive fixes applied to HCG 16's segmentation, in the order diagnosed. Each stage is scoped so that Stephan's Quintet and HCG 90 are unaffected unless noted.

Stage	Fix Applied	Outcome
8b	Raised significance threshold k_{hi}: $10\sigma \rightarrow 25\sigma$.	Foreground fraction barely moved, $9.88\% \rightarrow 9.53\%$ – confirmed the threshold itself was not the real cause.
8d	Compute a no-data mask from raw non-finite pixels *before* any fill; exclude it from local background/noise statistics and from the final significance test.	Foreground fraction dropped to 0.22% – the actual fix.
8e	Dilate the no-data mask by a tunable margin (`nodata_buffer_px`$=50$) before use.	Removed $\sim$19 of 28 remaining candidates, which were false detections strung along the mosaic's diagonal boundary, not real sources.
8f	Fold low-weight pixels (from the FITS WHT extension) into the no-data mask; expose `area_max_px` as a per-target override.	Removed 5 of 17 remaining false detections along a second, jagged drizzle-edge artifact, and corrected 2 objects (diffraction-spike/blooming stars) previously misclassified as galaxies.
8g	Replace the weight-mask's percentile-based reference with the histogram mode; add hard 70%/90% no-data safety ceilings.	Fixed a regression introduced by 8f itself, where an inflated percentile reference pushed total no-data to 98.8% and silently dropped HCG 16 from the run entirely.

3.3 Final Configuration

Table 2: HCG 16's accumulated per-target segmentation overrides at end of week. All other targets use the unmodified global defaults.

Parameter	Value
Significance threshold (k_{hi})	25.0
Minimum candidate area (px)	60
No-data mask buffer (px)	50
Maximum candidate area (px)	5000
Weight-mask fraction threshold	0.05
Per-object diagnostic printing	Enabled

Every value above is flagged in-pipeline as a first-attempt, evidence-informed starting point, not a validated final setting. The recurring lesson across stages 8f–8g is methodological as much as numerical: a percentile-based reference is fragile against rare elevated regions in real archival data, while a histogram-mode reference combined with an explicit safety ceiling degrades gracefully instead of failing silently.

4 Conclusion and Next Steps

Week 8 converted the pipeline from a single hard-coded target into a reusable multi-target architecture and brought HCG 90 and HCG 16 online under it, but the majority of the week's effort went into an unplanned, iterative diagnosis of HCG 16's extreme false-positive rate – a reminder that scaling a validated single-target pipeline to new archival data is rarely a drop-in operation. Two items are carried forward explicitly:

1. HCG 16's candidate list still contains likely cosmic-ray / satellite-streak artifacts, deliberately left unfiltered this week and flagged as a caveat on any downstream result that uses this target.
2. The decision-rule revision flagged in Week 7 (no upper asymmetry bound) was not addressed this week – it was deferred until the pooled, multi-target feature table from this week's work could inform a data-driven, rather than

purely literature-anchored, fix.

4.1 Work Planned for Week 9

- **Classification and accuracy benchmark**: apply the existing rule-based classifier to the full combined feature table and formally score accuracy against Stephan's Quintet's known members, the only target with real ground truth.
- **HCG 90 / HCG 16 reporting**: classify both new targets' candidates and report them clearly as unscored, since neither has a name-matching catalog to benchmark against.

End Of Weekly Report